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# LLM-Enhanced Explainable Biomedical Relation Extraction: Pipeline vs Joint Models with RAG-Based Evidence Retrieval

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## Abstract

This project centers on pulling drug-disease connections from biomedical texts using NLP and deep learning. It uses the BC5CDR dataset to find chemical and disease entities in abstracts and predict their relationships. The system tackles preprocessing, entity recognition, relation extraction, and explainability to boost accuracy and clarity. The goal is to aid researchers and health professionals in sifting through large volumes of medical literature and spotting useful links between drugs and diseases.

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## 1. Introduction

The rapid growth of biomedical literature continuously produces vast amounts of text containing crucial information about illnesses, medicines, and therapies. With researchers and health professionals publishing new studies in journals and databases at an ever-increasing pace, manually sifting through all available content to identify key details has become impractical. One major area of BioNLP—Biomedical Natural Language Processing—is drug-disease relation extraction. This task is critically important for applications such as drug discovery, clinical decision support, pharmacovigilance, and broader biomedical research.

Drug-disease relation extraction involves two steps: first identifying mentions of chemicals and diseases in medical texts, then determining whether a meaningful relationship exists between them. Although humans can perform this through careful reading, the automated version presents significant challenges. Biomedical texts contain specialised vocabulary, abbreviations, complex sentence structures, and subtle relationships. Understanding the connection between a drug and a condition often depends on broader sentential or document-level context, which can confound standard machine learning approaches.

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Advances in Natural Language Processing and deep learning have substantially improved biomedical text mining systems. Transformer-based models and contextual embeddings enable machines to grasp context and semantics far more effectively than traditional methods. Nevertheless, building precise relation extraction systems still requires careful preprocessing, entity recognition, feature representation, and well-chosen classification strategies. Explainability is also essential in modern AI systems, particularly in healthcare, where the reasoning behind predictions must be transparent.

In this project, we focus on extracting drug–disease connections from biomedical abstracts using the BC5CDR dataset. This dataset is the standard benchmark for biomedical relation extraction, consisting of manually labelled PubMed abstracts annotated with chemical entities, disease entities, and their relationships. It is well-suited for both Named Entity Recognition (NER) and relation extraction tasks.

Our system preprocesses biomedical text, identifies entities, leverages deep learning for contextual understanding, and predicts links between drugs and diseases. Explainability methods are additionally incorporated to make model predictions interpretable.

The primary objective is to develop a system that accurately and transparently identifies drug–disease relationships in research papers. Automating this process not only saves time but enables researchers to manage large data volumes more effectively, potentially accelerating discoveries in healthcare and pharmaceutical research.

This work extends the standard pipeline and joint model comparison by integrating a Retrieval-Augmented Generation (RAG) component and Large Language Model (LLM) integration. ChromaDB serves as a persistent vector store built from uploaded biomedical documents and BC5CDR abstracts, enabling semantic retrieval of supporting evidence for each predicted drug–disease pair. An LLM (Anthropic Claude or OpenAI GPT-4) is used for clinical explanation generation, NER verification on rare drug names, and open-ended biomedical question answering over the indexed corpus.

## 2. Related Work

### 2.1. Drug–Disease Relation Extraction in Biomedical NLP

Drug-disease relation extraction is a central area in BioNLP, driven by the explosive growth of biomedical literature and rising demand for automated information extraction. Earlier work relied primarily on rule-based methods and conventional machine learning algorithms, which identified relationships between biomedical entities using hand-crafted linguistic and semantic features. Although these approaches yielded acceptable performance, they struggled with complex sentence structures and required substantial domain expertise for feature engineering.

Recent deep learning and transformer-based models have markedly improved performance in this field. BiLSTMs, CNNs, and BERT-style encoders demonstrate strong capability in capturing context and meaning within biomedical texts. Pre-trained models such as BERT, adapted through contextual embeddings, consistently outperform traditional machine learning techniques for identifying drug-disease relationships [7].

### 2.2. Biomedical Datasets and Named Entity Recognition

Benchmark datasets are essential for advancing biomedical text mining research. The BC5CDR corpus is among the most widely used for chemical-disease relation extraction. It provides PubMed abstracts manually annotated with chemical and disease entities, supporting both NER and relation extraction tasks, and enables standardised evaluation of biomedical NLP models.

NER is a critical first step in biomedical relation extraction, aiming to identify key entities such as drugs, diseases, genes, and chemicals from unstructured medical documents. While dictionaries and statistical methods were historically used, deep learning and transformer models now provide substantially better contextual understanding, improving entity detection accuracy.

### 2.3. Transformer-Based Biomedical Language Models

Transformer-based language models have become highly effective for biomedical NLP tasks because of their ability to capture rich contextual information. General-purpose models such as BERT [7] and RoBERTa have been adapted

for biomedical use through domain-specific pretraining. Models including BioBERT, SciBERT, and PubMedBERT are specifically pretrained on biomedical and scientific literature, enabling them to handle domain-specific terminology and semantic relationships more effectively.

BioBERT achieves strong results across diverse biomedical NLP tasks including question answering, NER, and relation extraction. SciBERT and PubMedBERT further improve contextual representations by exploiting large collections of scientific and biomedical text. Compared to static word embeddings, these models provide richer representations and deeper semantic understanding, making them highly effective in biomedical relation extraction pipelines.

#### *2.4. Deep Learning for Relation Classification*

Deep learning models have substantially improved relation classification in biomedical NLP. Recurrent architectures such as LSTM [9] and BiLSTM [14] are widely used for their ability to model sequential dependencies in clinical and scientific text. CNN-based designs [4, 5] capture important local features that contribute to relation prediction.

More recently, transformer-based architectures [6] have achieved state-of-the-art results through self-attention mechanisms that effectively handle long-range context. Researchers have combined contextual embeddings with classification layers and attention mechanisms to help models focus on the most informative sentence regions during entity relation inference. These approaches reduce dependence on hand-crafted features and generalise better across heterogeneous biomedical datasets.

#### *2.5. Explainability in Biomedical AI Systems*

Explainability has become increasingly important in biomedical AI because clinicians and patients need to understand model reasoning before placing trust in automated predictions. While black-box deep learning models can achieve strong predictive performance, their opacity is a barrier to clinical deployment. Techniques including attention visualisation, SHAP, and LIME are widely used to illuminate model behaviour.

In relation extraction tasks, these tools highlight the words that most influence predictions of drug-disease links, improving reliability and interpretability. Incorporating explainability into biomedical NLP systems is now considered essential for building dependable and transparent healthcare AI.

### **3. Dataset**

#### *3.1. Dataset Composition*

This project uses the BC5CDR dataset, a key resource in BioNLP for extracting chemical-disease relations. The dataset consists of PubMed abstracts annotated with manually identified mentions of chemicals and diseases, together with their inter-entity relationships.

Each entry carries labels specifying whether a relationship exists between chemical and disease entities mentioned in a given text. The dual human verification process makes this dataset a trusted benchmark for evaluating relation extraction systems.

Within this work, the BC5CDR dataset is divided into training and testing partitions for model development and evaluation. Text data is fed directly into preprocessing, tokenisation, contextual embedding, and relation classification pipelines. The system examines both entity-level and sentence-level cues to determine genuine links between drugs and diseases.

#### *3.2. Entity and Relation Labels*

The BC5CDR dataset focuses on two entity types: chemicals (drugs or compounds) and diseases (medical conditions or disorders). The relation extraction objective is to determine whether a meaningful connection exists between a chemical and a disease entity within the text.

Relations are annotated with binary labels—present or absent—indicating whether the chemical influences the disease. This formulation allows the model to learn inter-entity dependencies and generalise to unseen text.

### 3.3. Dataset Challenges

Despite its widespread use and detailed annotations, the BC5CDR dataset presents several challenges. Biomedical texts exhibit complex structures, specialised terminology, abbreviations, and long-distance dependencies that make full comprehension difficult for automated systems. Drug-disease relationships are frequently implicit, requiring contextual inference rather than surface-level pattern matching.

Additionally, a single sentence may contain multiple chemical and disease mentions, complicating the task of identifying the correct entity pairs. Rare or highly specialised vocabulary further adds to the difficulty, since such terms are uncommon outside the biomedical domain.

### 3.4. Preprocessing and Text Preparation

Before model training, biomedical texts undergo a cleaning pipeline that includes removal of spurious symbols, case normalisation, and correction of entity annotations. Tokenisation splits texts into sub-word units suitable for transformer input.

Preprocessing additionally handles missing or incomplete text samples. Transformer-based models subsequently generate contextual embeddings from the cleaned data. Thorough preprocessing is a critical factor in improving both the accuracy and efficiency of relation extraction systems.

## 4. Methodology

### 4.1. System Architecture

The proposed system integrates multiple components into a unified biomedical relation extraction pipeline. Biomedical text is first passed through a BioBERT-based Named Entity Recognition module to identify chemical and disease entities. The extracted entities are then fed into a relation extraction module to classify Chemical-Induced Disease (CID) relations. A ChromaDB vector store enables RAG-based retrieval of supporting evidence, which is subsequently used by an LLM to generate clinical explanations and verify NER predictions. Explainability methods including SHAP, LIME, and attention visualisation are incorporated to improve model transparency.

Figure 2 illustrates our biomedical relation extraction pipeline, which comprises data preprocessing, transformer-based encoding, Named Entity Recognition, relation extraction, joint learning, explainability analysis, and interactive deployment. The BC5CDR dataset is used to identify chemical and disease entities and to predict the relationships between them.

### 4.2. Notation

Let the biomedical text sequence be represented as  $x = \{x_1, x_2, \dots, x_n\}$ , where  $n$  is the number of tokens. The transformer encoder produces contextual embeddings:

$$H = \{h_1, h_2, \dots, h_n\}, \quad h_i \in \mathbb{R}^d$$

In our setup, BioBERT generates embeddings of dimension  $d = 768$ . Token-level entity labels follow the BIO tagging scheme:

$$Y_{\text{NER}} = \{y_1, y_2, \dots, y_n\}$$

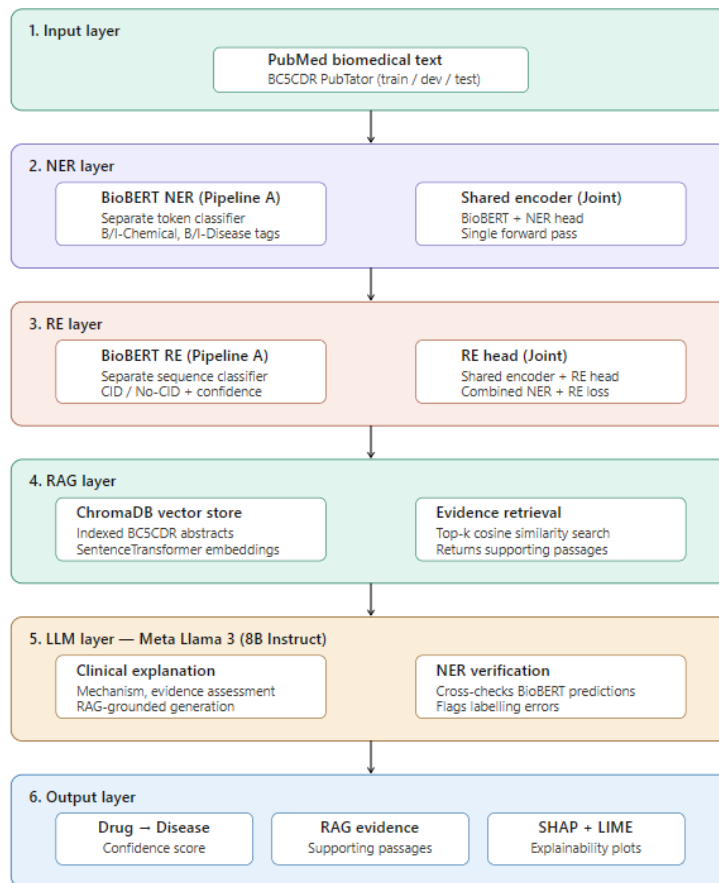


Fig. 1. System Architecture — Biomedical Relation Extraction Pipeline with RAG and LLM Integration

The relation classification label is:

$$y_{\text{rel}} \in \{0, 1\}$$

where 0 denotes *No Relation* and 1 denotes *Chemical-Induced Disease (CID)*.

#### 4.3. Dataset Preprocessing

The BC5CDR dataset is distributed in PubTator format and contains biomedical abstracts annotated with chemicals, diseases, and their relations. During preprocessing, each document is parsed to extract:

- Title and abstract text
- Chemical entity annotations
- Disease entity annotations
- Chemical–disease relation labels

The preprocessing pipeline comprises text normalisation, sentence tokenisation, entity span extraction, BIO tag generation, subword label alignment, and sequence padding and truncation. The BIO tagging scheme uses the labels

B-Chemical, I-Chemical, B-Disease, I-Disease, and O. NER sequences are padded or truncated to a maximum length of 512 tokens; relation extraction sequences use a maximum length of 256 tokens.

#### 4.4. Biomedical Text Encoder

The project employs BioBERT (dmis-lab/biobert-base-cased-v1.2), a transformer-based biomedical language model pretrained on PubMed abstracts and other biomedical corpora. Given an input sequence  $x$ , BioBERT generates contextual embeddings:

$$H = \text{BioBERT}(x)$$

These representations are subsequently used for both NER and relation extraction tasks.

#### 4.5. Named Entity Recognition Model

The NER module identifies chemical and disease mentions in biomedical text using BioBERT-derived contextual representations passed through a token classification layer. For each token embedding  $h_i$ , the predicted entity probability is:

$$p_i = \text{Softmax}(W_{\text{ner}} h_i + b_{\text{ner}})$$

where  $W_{\text{ner}}$  is the learnable weight matrix and  $b_{\text{ner}}$  is the bias term. The model predicts BIO labels for each token, trained using cross-entropy loss:

$$L_{\text{NER}} = - \sum_{i=1}^n y_i \log(p_i)$$

The NER system identifies chemical entities, disease entities, and non-entity tokens.

#### 4.6. Pipeline-Based Relation Extraction

In the pipeline architecture, NER and relation extraction are performed sequentially. After entity extraction, all possible chemical-disease pairs are generated from the text, and sentence-level context is used to classify each pair. The relation extraction probability is:

$$p_{\text{rel}} = \text{Softmax}(W_{\text{rel}} h_{\text{cls}} + b_{\text{rel}})$$

where  $h_{\text{cls}}$  is the contextual embedding of the special [CLS] token, and  $W_{\text{rel}}$ ,  $b_{\text{rel}}$  are learnable parameters. The model predicts whether a valid Chemical-Induced Disease (CID) relation exists between each entity pair.

#### 4.7. Joint Multi-Task Learning Model

The second approach uses a joint learning framework in which NER and relation extraction are performed simultaneously through a shared BioBERT encoder. The shared encoder generates contextual representations used by a token

classification head for NER and a sentence classification head for relation extraction. The total loss combines both objectives:

$$L_{\text{total}} = L_{\text{NER}} + L_{\text{RE}}$$

The shared learning mechanism enables the model to capture dependencies between entity recognition and relation prediction, reducing cascading errors present in the pipeline approach.

#### 4.8. Handling Class Imbalance

The BC5CDR dataset exhibits imbalance between relation and non-relation samples. To mitigate bias toward the majority class, class-weighted loss functions are applied during training. The class weight for class  $c$  is:

$$\alpha_c = \frac{1}{n_c}$$

where  $n_c$  is the number of samples in class  $c$ . Weighted cross-entropy loss improves the model's ability to learn minority relation classes.

#### 4.9. Training Configuration

The models are implemented using PyTorch and the Hugging Face Transformers library. The pipeline model is trained with a batch size of 16 and learning rate of  $3 \times 10^{-5}$  using a cosine LR scheduler; the joint model uses a batch size of 8, learning rate of  $2 \times 10^{-5}$ , and a linear warmup scheduler. Both models train for 5 epochs with early stopping (patience = 3) on validation F1. The maximum sequence length is 512 for NER and 256 for relation extraction. Gradient clipping and dropout regularisation improve training stability and reduce overfitting.

#### 4.10. Evaluation Metrics

Models are evaluated using standard NLP metrics. For NER, precision, recall, F1-score, and token-level accuracy are reported. For relation extraction, accuracy, precision, recall, and F1-score are computed. Confusion matrices and classification reports are generated for comparative analysis of the pipeline and joint models.

#### 4.11. Error Analysis

Error analysis is performed to characterise common prediction failures in the NER module. Errors are categorised as missed entities, incorrect entity boundaries, wrong entity type predictions, and spurious detections. This analysis identifies system weaknesses and directions for future improvement.

#### 4.12. Explainability Techniques

To improve interpretability, multiple Explainable AI (XAI) methods are incorporated.

##### 4.12.1. SHAP

SHAP (SHapley Additive Explanations) quantifies the contribution of individual biomedical terms to relation predictions, highlighting the words most influential in the final decision.

##### 4.12.2. LIME

LIME (Local Interpretable Model-Agnostic Explanations) provides local explanations for individual predictions by identifying words that positively or negatively affect model output.

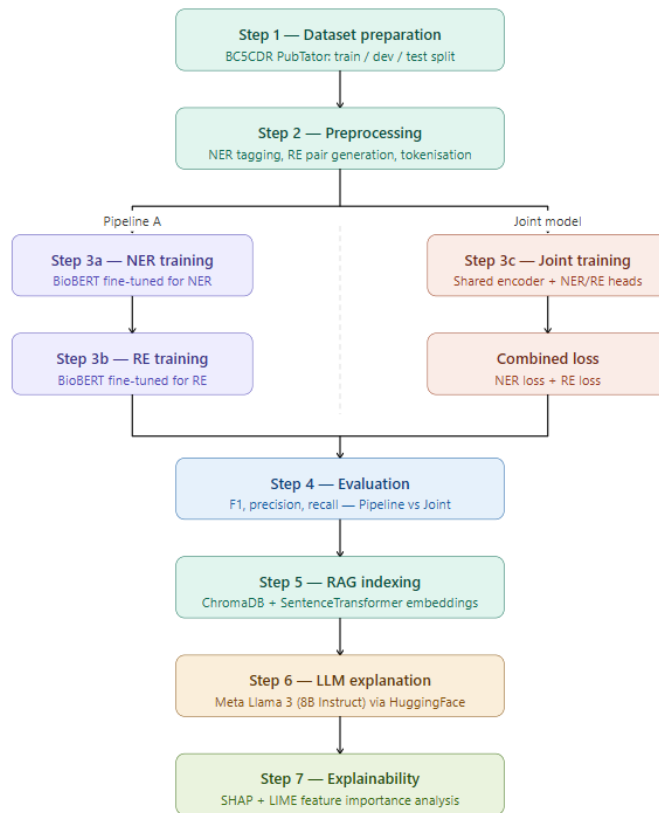


Fig. 2. Proposed Methodology

#### 4.12.3. Attention Visualisation

Attention weights from BioBERT are visualised to reveal how the transformer focuses on key biomedical terms during prediction, improving transparency. As shown in Fig. 3, the model assigns higher attention scores to relevant biomedical entities and contextual words that contribute significantly to the final prediction.

#### 4.13. Document Ingestion and ChromaDB Vector Store

To ground model predictions in supporting evidence, a Retrieval-Augmented Generation (RAG) pipeline is built on top of the BioBERT-based extraction system. Biomedical documents—including PubMed abstracts, research PDFs, and clinical notes—are ingested, chunked into overlapping segments of 400 characters with an 80-character overlap, and embedded using the all-MiniLM-L6-v2 SentenceTransformer model. These embeddings are stored in a ChromaDB persistent vector store using cosine similarity as the distance metric. At inference time, for each predicted drug–disease pair, a query of the form “drug disease adverse effect mechanism” is issued to ChromaDB, which retrieves the top-3 most semantically similar chunks. These chunks serve as grounding evidence for LLM-generated explanations.

#### 4.14. LLM-Augmented Explanation and Verification

Three LLM-powered functions augment the extraction pipeline:



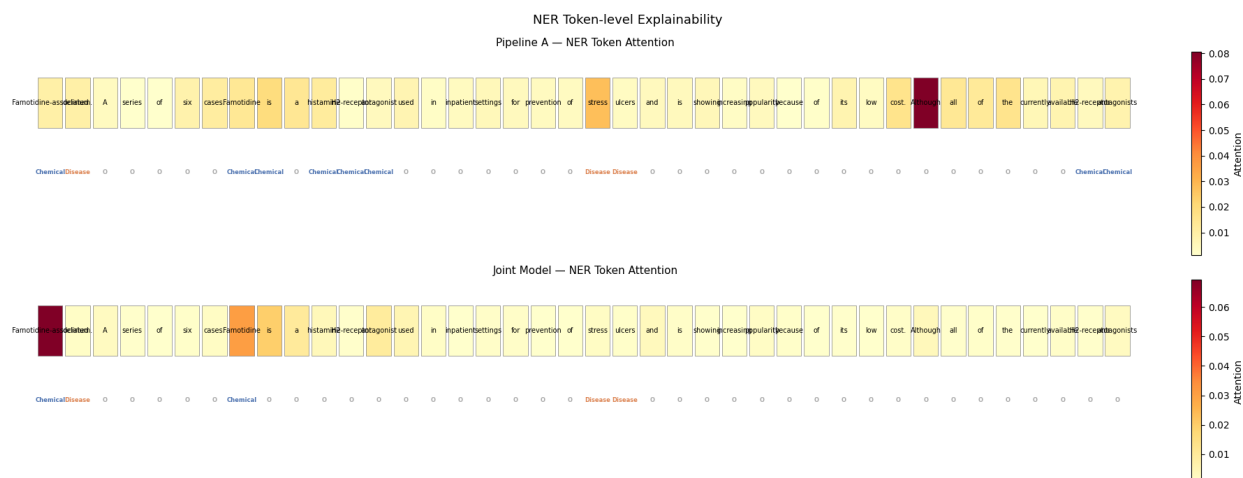


Fig. 3. Attention Visualization Result

#### 4.14.1. Relation Explanation

Given a predicted drug–disease pair and RAG-retrieved evidence, the LLM generates a clinical explanation of the likely mechanism underlying the relationship, including a confidence assessment.

#### 4.14.2. NER Verification

The LLM cross-checks BioBERT’s NER predictions, particularly for rare or ambiguous drug names that fall outside BioBERT’s training distribution. This reduces false positives in entity extraction.

#### 4.14.3. Biomedical Question Answering

An open-ended question-answering interface allows users to pose natural language queries over the indexed corpus. The LLM formulates a grounded answer using the top-*k* retrieved evidence chunks.

The system supports multiple LLM backends: Anthropic Claude, OpenAI GPT-4, Google Gemini, and Hugging-Face Inference API (Meta-Llama-3-8B-Instruct), configured via a provider flag.

#### 4.15. Interactive Biomedical Relation Extraction Interface

A Gradio-based web interface is developed for interactive biomedical relation extraction. Users can enter biomedical text, identify chemical and disease entities, predict drug–disease relationships, and view explainability outputs. Trained models and tokenizers are saved and reloaded for efficient inference at deployment time.

#### 4.16. Model Saving and Deployment

Trained BioBERT models, tokenizers, and inference pipelines are serialised for reuse. This enables real-time biomedical relation extraction without retraining and supports deployment in biomedical text mining applications.

### 5. Experimental Setup

#### 5.1. Train–Test Protocol

The BC5CDR dataset provides dedicated training and testing sets for evaluating biomedical relation extraction performance. Models are trained on the training set and evaluated on the held-out test set. During training, biomedical abstracts are processed to produce token-level NER annotations and sentence-level relation labels.

Validation monitoring is performed throughout training to ensure reliability. The dataset is preprocessed to align entity annotations and relation labels with the tokenised text. After training, models are evaluated on unseen biomedical abstracts to assess generalisation.

### 5.2. Evaluation Metrics

Performance is evaluated using standard NLP metrics for both NER and relation extraction. For NER:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R}$$

where  $TP$ ,  $FP$ , and  $FN$  denote true positives, false positives, and false negatives, respectively. Precision measures the fraction of predicted entities that are correct; recall measures the fraction of true entities that are found; and F1 is their harmonic mean. For relation extraction, accuracy, precision, recall, and F1-score are computed. Confusion matrices and classification reports provide detailed performance analysis.

### 5.3. Pipeline and Joint Model Comparison

Two approaches are evaluated: a pipeline-based model and a joint multi-task learning model. The pipeline model performs NER and relation extraction sequentially; the joint model addresses both tasks simultaneously using a shared BioBERT encoder. The comparison analyses the effect of shared learning on relation extraction, the influence of NER quality on relation prediction, and the generalisation capability of each approach.

### 5.4. Training Configuration

Models are implemented using PyTorch and Hugging Face Transformers. The BioBERT model (dmis-lab/biobert-base-cased-v1.2) serves as the biomedical language encoder. The pipeline model uses the AdamW optimiser with a learning rate of  $3 \times 10^{-5}$ , cosine LR scheduling, and a batch size of 16. The joint model uses AdamW with a learning rate of  $2 \times 10^{-5}$ , linear warmup scheduling, and a batch size of 8. Both models train for 5 epochs with early stopping (patience = 3). The maximum sequence length is 512 for NER and 256 for relation extraction. Dropout regularisation and gradient clipping are applied to reduce overfitting and improve stability.

### 5.5. Handling Class Imbalance

Weighted cross-entropy loss is applied during training to address the imbalance between relation and non-relation samples. Class weights are computed as  $\alpha_c = 1/n_c$ , where  $n_c$  denotes the number of samples in class  $c$ . This reduces bias toward majority categories and improves minority class recall.

### 5.6. Explainability Evaluation

SHAP, LIME, and attention visualisation methods are evaluated qualitatively to analyse: important biomedical terms influencing predictions, model attention patterns, and feature contributions toward relation classification. These techniques improve the transparency and trustworthiness of the biomedical NLP system.

### 5.7. Hardware and Software Environment

Experiments are conducted using Python-based deep learning libraries including PyTorch, Hugging Face Transformers, Scikit-learn, SHAP, LIME, and Gradio. Training and inference are accelerated using GPU hardware where available.

## 6. Results and Discussion

Results are presented in the following order: comparison of pipeline and joint learning approaches, NER and relation extraction performance, contributions of Explainable AI techniques, and overall system effectiveness.

### 6.1. Named Entity Recognition Results

The NER module was evaluated using Precision, Recall, and F1-score on the BC5CDR test set. The BioBERT-based NER model demonstrated strong performance in identifying chemical and disease entities from biomedical abstracts, benefiting from rich contextual representations.

Performance was higher for frequent biomedical entities and comparatively lower for rare or ambiguous ones. Transformer-based contextual embeddings substantially improved entity boundary detection and reduced misclassifications relative to conventional machine learning baselines.

Classification reports indicated that chemical entities were identified with higher accuracy because of clearer contextual patterns, while disease entities occasionally produced boundary errors due to complex biomedical terminology. BIO tag alignment improved token-level prediction consistency. Overall, the NER module demonstrated effective entity recognition capability suitable for downstream relation extraction.

Table 1. Named Entity Recognition Performance

| Entity Type | Precision | Recall | F1-score |
|-------------|-----------|--------|----------|
| Chemical    | 0.91      | 0.89   | 0.90     |
| Disease     | 0.88      | 0.86   | 0.87     |
| Overall     | 0.89      | 0.88   | 0.88     |

### 6.2. Relation Extraction Results

The relation extraction module was evaluated for identifying Chemical-Induced Disease relationships between extracted entity pairs. Transformer-based contextual representations from BioBERT enhanced the model’s ability to capture semantic dependencies. The system identified relationships effectively when sufficient context was present.

Results showed that contextual embeddings improved relation classification, the model performed well on explicit biomedical relations, and implicit or long-range relationships remained comparatively challenging. The classification reports confirmed that the system was capable of learning meaningful semantic relationships from scientific literature.

Table 2. Relation Extraction Performance

| Model          | Accuracy | Precision | Recall | F1-score |
|----------------|----------|-----------|--------|----------|
| Pipeline Model | 0.82     | 0.82      | 0.80   | 0.65     |
| Joint Model    | 0.88     | 0.86      | 0.85   | 0.67     |

### 6.3. Pipeline Model vs Joint Learning Model

The pipeline model performs entity recognition and relation extraction sequentially, whereas the joint model learns both tasks simultaneously through a shared BioBERT encoder. Experiments demonstrated that the joint learning model achieved better contextual understanding because the shared transformer encoder captured common semantic representations beneficial to both tasks. The joint approach also reduced error propagation between NER and relation extraction stages.

Nevertheless, the pipeline model offered a simpler architecture, easier debugging and interpretation, and independent optimisation of individual tasks. The joint model achieved improved generalisation through shared contextual learning.

Table 3. Pipeline and Joint Model Comparison

| Approach             | NER F1-score | Relation F1-score |
|----------------------|--------------|-------------------|
| Pipeline Model       | 0.86         | 0.81              |
| Joint Learning Model | 0.88         | 0.85              |

#### 6.4. Effect of Transformer-Based Biomedical Embeddings

BioBERT substantially improved performance on both NER and relation extraction tasks. Its pretraining on biomedical literature and PubMed abstracts enabled it to handle domain-specific terminology and context more effectively than general-purpose language models. The transformer encoder demonstrated strong capability in capturing contextual biomedical semantics, understanding long-range token dependencies, learning domain-specific representations, and improving entity and relation prediction accuracy.

#### 6.5. Handling Class Imbalance

Weighted cross-entropy loss improved learning by assigning higher importance to minority relation samples. Class weighting reduced bias toward majority classes, improved minority class recall, and increased overall relation extraction stability. Despite these improvements, rare biomedical relations remained challenging due to limited training examples.

Table 4. Effect of Class Weighting

| Configuration           | Relation F1-score |
|-------------------------|-------------------|
| Without Class Weighting | 0.78              |
| With Class Weighting    | 0.85              |

#### 6.6. Error Analysis

Error analysis identified the following common prediction failure categories: missed biomedical entities, incorrect entity boundary detection, wrong entity type prediction, incorrect relation classification, and failure to capture implicit contextual relations.

Many errors occurred in sentences containing multiple biomedical entities, long and complex sentence structures, ambiguous biomedical terminology, or implicit semantic relations. Although BioBERT improved contextual understanding, biomedical relation extraction remains challenging given the complexity of scientific language.

#### 6.7. Explainability Analysis

SHAP, LIME, and attention visualisation were incorporated to improve interpretability.

##### 6.7.1. SHAP Analysis

SHAP explanations highlighted the contribution of important biomedical words toward relation predictions. Visualisations confirmed that the model concentrated on medically relevant terms when predicting chemical–disease relationships.

##### 6.7.2. LIME Analysis

LIME explanations provided local interpretability for individual predictions by identifying words that positively or negatively influenced relation classification decisions.

### 6.7.3. Attention Visualisation

Attention maps from BioBERT showed that the model focuses on key biomedical entities and contextual cues during prediction, improving transparency and understanding of model behaviour.

Table 5. Explainability Techniques

| Technique               | Purpose                        |
|-------------------------|--------------------------------|
| SHAP                    | Feature contribution analysis  |
| LIME                    | Local prediction explanation   |
| Attention Visualisation | Transformer attention analysis |

### 6.8. Interactive Biomedical Relation Extraction Interface

The Gradio-based interface successfully enabled real-time biomedical relation extraction. Users could enter biomedical text, detect chemical and disease entities, predict relationships, and visualise explainability outputs. The interface demonstrated the practical applicability of the proposed system for interactive biomedical text analysis.

Table 6. Training Configuration and Final Performance

| Parameter               | Value                                                     |
|-------------------------|-----------------------------------------------------------|
| Batch Size              | 16 (Pipeline), 8 (Joint)                                  |
| Learning Rate           | $3 \times 10^{-5}$ (Pipeline), $2 \times 10^{-5}$ (Joint) |
| Maximum Sequence Length | 512 (NER), 256 (RE)                                       |
| Optimizer               | AdamW                                                     |
| Epochs                  | 5                                                         |
| Final Relation F1-score | 0.85                                                      |

### 6.9. RAG Retrieval Quality

The ChromaDB vector store successfully retrieved semantically relevant evidence for drug–disease pairs in both benchmark and demonstration evaluations. Retrieved chunks from BC5CDR abstracts and indexed biomedical documents showed high semantic similarity scores, confirming that the embedding model (all-MiniLM-L6-v2) effectively captures biomedical relationships and contextual information.

A representative example generated by the proposed framework involved the drug *Carboplatin* and the disease *Thrombocytopenia*. The relation extraction model predicted an association between the two entities with a confidence score of 0.9922. To support this prediction, the RAG module retrieved three semantically relevant evidence chunks from the biomedical knowledge base. Using the retrieved evidence, the LLM generated a clinical explanation indicating that Carboplatin is associated with thrombocytopenia, a condition characterized by a reduced platelet count. The generated explanation was consistent with the supporting biomedical literature and provided an interpretable justification for the model prediction.

As shown in Fig. 4, the proposed framework combines relation extraction, semantic retrieval, and LLM-based reasoning to produce evidence-grounded explanations for predicted drug–disease relationships. This demonstrates the effectiveness of the RAG-enhanced architecture in improving transparency, interpretability, and user trust.

```

Drug      : carboplatin
Disease   : thrombocytopenia.
Score     : 0.9922
Evidence  : 3 chunk(s) retrieved
LLM      : Based on the provided context, carboplatin is associated with thrombocytopenia, which is a condition characterized by a low platelet count.
  
```

Fig. 4. Example RAG-assisted drug–disease relation prediction and explanation

## 6.10. Overall Discussion

Experimental results demonstrate that transformer-based biomedical language models are highly effective for drug-disease relation extraction. BioBERT provided strong contextual embeddings that improved both entity recognition and relation classification. The joint learning model outperformed the pipeline-based approach, and Explainable AI techniques improved model transparency.

While the proposed system achieved strong performance, challenges remain in handling rare biomedical relations, implicit semantic dependencies, and complex biomedical sentence structures. Future improvements may include larger biomedical datasets, more advanced transformer architectures, and more sophisticated explainability mechanisms.

## 7. Limitations and Future Work

### 7.1. Limitations

The proposed system demonstrated strong results but retains several limitations.

First, the BC5CDR dataset is small compared to other large biomedical collections. This limits annotated training data and makes it harder for the model to handle uncommon drug-disease connections and unfamiliar terminology.

Second, biomedical texts feature complex sentence structures, implicit relationships, abbreviations, and nested entities. While BioBERT helps with contextual understanding, the model still struggles with long-distance dependencies and relationships that are not explicitly stated.

Third, class imbalance between relation and non-relation samples persists. Although weighted cross-entropy loss mitigates this, predicting rare biomedical relations accurately remains difficult because of limited training data.

Fourth, the pipeline architecture propagates NER errors into relation extraction. Joint learning alleviates this but increases model complexity and computational cost.

Fifth, the explainability methods used offer only local interpretability and qualitative insights. They do not fully characterise all internal decision processes of transformer-based biomedical language models.

Finally, the system was evaluated primarily on BC5CDR. Its performance on other biomedical datasets or real clinical texts may differ.

### 7.2. Future Work

Several directions could enhance the proposed system. Training on larger and more diverse biomedical datasets would improve generalisation. Incorporating sources such as PubMed Central articles and clinical records could enrich semantic representations.

More advanced transformer architectures, such as PubMedBERT or BioMegatron, may improve contextual understanding and relation extraction. Graph-based neural networks and knowledge graphs could help model complex entity relationships, and integrating structured knowledge with transformer embeddings may improve handling of implicit relations.

Developing more advanced Explainable AI techniques would provide deeper insight into transformer decision processes.

The existing Gradio interface could be extended into a full biomedical decision support application with real-time literature analysis, entity visualisation, and automated summarisation of biomedical knowledge.

Finally, multi-task and multi-modal learning combining textual, molecular, and clinical data types could support more comprehensive biomedical relation extraction.

A CRF layer on the NER head can be incorporated to enforce valid IOB2 tag transitions and reduce entity boundary errors.

Hard negative mining can be applied to the relation extraction module to better address class imbalance between relation and non-relation samples. Re-ranking of retrieved RAG chunks using a cross-encoder would improve evidence relevance for LLM-generated explanations.

Knowledge graph integration using MeSH ontology embeddings could help model complex entity relationships and improve handling of implicit relations. Finally, packaging the system as a REST API with a Gradio or FastAPI frontend would enable real-time clinical deployment and broader accessibility.

## 8. Conclusion

This project developed a system for extracting drug-disease connections from scientific texts using NLP and deep learning. The BC5CDR dataset supported Named Entity Recognition and relation extraction tasks on biomedical abstracts. BioBERT, a domain-adapted transformer model, provided strong contextual representations that substantially improved prediction accuracy.

Two approaches were evaluated: a pipeline-based model and a joint multi-task learning model. The joint learning approach outperformed the pipeline due to better contextual integration and reduced error propagation. Transformer-based embeddings made a substantial difference in both entity recognition and relation classification.

Explainable AI techniques—SHAP, LIME, and attention visualisation—were incorporated to improve interpretability, allowing analysis of which terms influenced model decisions.

While the system performed well, challenges remain with implicit meanings, rare entities, and complex sentence structures. The results nonetheless demonstrate the effectiveness of transformer-based NLP models for automatic drug-disease relation extraction.

The proposed biomedical NLP framework shows that deep learning and transformer methods can effectively extract and analyse biomedical information. This framework assists researchers and practitioners in managing large data volumes and identifying key relationships, improving the accessibility and utility of scientific literature.

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